

TAEYOUNG YUN
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in Taeyoung Yun

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RESEARCH INTEREST

My research interests lie in controllable generative modeling with large models (e.g., LLMs, Diffusion / Flow-based models) by exploring their latent spaces. In particular, I'm interested in building an amortized sampler that can extract crucial latents to generate desired samples. To accomplish this, my research focuses on amortized inference for generative models using off-policy reinforcement learning (RL) methods. I've also participated in several transportation-related projects based on high-dimensional black-box optimization methods. My current research topics are:

- **RL for LLMs:** Developing an off-policy RL algorithm for post-training of LLMs to solve diverse downstream tasks (**[IC7], [IC13]**).
- **GFlowNets for Scientific Discovery:** Developing an off-policy RL algorithm called GFlowNets to search high-performing and diverse candidates for scientific discovery (**[IC1], [C2], [C6], [C8], [P3]**).
- **Generative models for Optimization:** Developing a generative model-based approach (Diffusion / Flow-based models) for searching promising candidates in high-dimensional space (**[IC9], [IC10], [IC11], [P1]**).

EDUCATION

03/2024 - Current	Ph.D Student in Industrial and Systems Engineering Supervised by Jinkyoo Park	KAIST
08/2022 - 02/2024	M.S in Graduate School of AI Supervised by Jinkyoo Park MS Thesis: Offline Meta Black-box Optimization Framework for Intelligent Traffic Light Management System	KAIST
03/2018 - 08/2022	B.S in Industrial and Systems Engineering & Computer Science	KAIST

INTERNSHIPS

06/2025 - 11/2025	Research Intern at Mila Hosted by Yoshua Bengio RL-based red-teaming with evolving environments by safety fine-tuning the victim LLM to promote easy-to-hard exploration.	Montreal, Canada
09/2024 - 03/2025	Visiting Intern at HKUST Hosted by Ling Pan Fine-tuning LLM with GFlowNets to generate diverse and effective prompts for text-to-image diffusion models.	Hong Kong (Remote)
03/2021 - 08/2021	Research Intern at Kakao Recommendation Team Develop contextual bandit algorithms for a personal recommendation. Analyze the gap between simulation and real-world deployment.	Seoul, Korea

INDUSTRIAL PROJECTS

03/2026 - Current	Agentic AI for Traffic-related Question Answering Collaborate with GS Develop an Agentic AI system for developing LLM-based for better traffic-related question answering framework.	Seoul, Korea
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09/2024 - 09/2025	Traffic Network Layout Optimization	Seoul, Korea
	Collaborate with GS Develop a Generative model-based design algorithm for optimizing traffic network layout on a given traffic pattern.	
03/2023 - 03/2024	Incentive Design for Managing Taxi Fleet	Daejeon, Korea
	Collaborate with ETRI Develop an RL-based incentive design algorithm for rebalancing taxi fleets to resolve the taxi imbalance problem.	
03/2022 - 03/2023	Traffic Light Optimization	Seoul, Korea
	Collaborate with KT Develop a Bayesian optimization algorithm for managing multiple traffic lights in the real world to reduce congestion.	

PUBLICATIONS

*: Equal Contribution

Preprints

Arxiv, 2026	[P4] Automated Kernel Discovery for Understanding High-dimensional Bayesian Optimization <u>Taeyoung Yun*</u> , Woocheol Shin*, Inhyuck Song, and Jinkyoo Park Paper / Code
Arxiv, 2026	[P3] Robust Exploration through Generative Replay Inhyuck Song, <u>Taeyoung Yun</u> , Jaewoo Lee, and Jinkyoo Park Paper / Code
Arxiv, 2025	[P2] Improving Sampling Distribution of Off-policy Training in Generative Flow Networks <u>Taeyoung Yun</u> , Sujin Yun, Jinkyoo Park, and Ling Pan Paper / Code
Arxiv, 2025 (based on NIPSW)	[P1] Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization Kiyong Om*, Kyuil Sim*, <u>Taeyoung Yun*</u> , Hyeongyu Kang, and Jinkyoo Park Paper / Code

Conference Publications

ICML, 2026	[IC13] Active Attacks: Red-teaming LLMs via Adaptive Environments <u>Taeyoung Yun</u> , Pierre-Luc St-Charles, Jinkyoo Park, Yoshua Bengio, and Minsu Kim Paper / Code
ICLR, 2026	[IC12] Diffusion Alignment as Variational Expectation-Maximization Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin, <u>Taeyoung Yun</u> , Kiyong Om, and Jinkyoo Park Paper / Code
WSDM, 2026	[IC11] Urban Traffic Network Layout Optimization with Guided Discrete Diffusion Models <u>Taeyoung Yun</u> , Inhyuck Song, Woocheol Shin, Yujin Shin, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park Paper / Code
KDD, 2025	[IC10] Wind Farm Layout Optimization with Diffusion Models Yujin Shin*, <u>Taeyoung Yun*</u> , Sujin Yun, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park Paper / Code

ICML, 2025 (based on ICLRW)	[C9] Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization <u>Taeyoung Yun</u> [*] , Kiyoung Om [*] , Jaewoo Lee, Sujin Yun, and Jinkyoo Park Paper / Code
ICML, 2025 (based on NIPSW)	[C8] Improved Off-Policy Reinforcement Learning in Biological Sequence Design Hyeonah Kim, Minsu Kim, <u>Taeyoung Yun</u> , Sanghyeok Choi, Emmanuel Bengio, Alex Hernández Garcia, and Jinkyoo Park Paper / Code
CVPR, 2025	[C7] Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation <u>Taeyoung Yun</u> , Dinghuai Zhang, Jinkyoo Park, and Ling Pan Paper / Code
ICLR, 2025	[C6] Adaptive Teachers for Amortized Samplers Minsu Kim [*] , Sanghyeok Choi [*] , <u>Taeyoung Yun</u> , Emmanuel Bengio, Leo Feng, Jarrod Rector-Brooks, Sungsoo Ahn, Jinkyoo Park, Nikolay Malkin, and Yoshua Bengio Paper / Code
NeurIPS, 2024	[C5] Guided Trajectory Generation with Diffusion Models for Offline Model-based Optimization <u>Taeyoung Yun</u> , Sujin Yun, Jaewoo Lee, and Jinkyoo Park Paper / Code
NeurIPS, 2024 (based on ICLRW)	[C4] GTA: Generative Trajectory Augmentation with Guidance for Offline Reinforcement Learning Jaewoo Lee [*] , Sujin Yun [*] , <u>Taeyoung Yun</u> , and Jinkyoo Park Paper / Code
KDD, 2024	[C3] An Offline Meta Black-box Optimization Framework for Adaptive Design of Urban Traffic Light Management Systems <u>Taeyoung Yun</u> [*] , Kanghoon Lee [*] , Sujin Yun, Ilmyung Kim, Won-Woo Jung, Min-Cheol Kwon, Kyujin Choi, Yoohyeon Lee, and Jinkyoo Park Paper / Code
ICML, 2024 (based on NIPSW)	[C2] Learning to Scale Logits for Temperature-conditional GFlowNets Minsu Kim [*] , Juhwan Ko [*] , <u>Taeyoung Yun</u> [*] , Dinghuai Zhang, Ling Pan, Woochang Kim, Jinkyoo Park, and Yoshua Bengio Paper / Code
ICLR, 2024 (Spotlight)	[C1] Local Search GFlowNets Minsu Kim, <u>Taeyoung Yun</u> , Emmanuel Bengio, Dinghuai Zhang, Yoshua Bengio, Sungsoo Ahn, and Jinkyoo Park Paper / Code

Journal Publications

TMLR, 2026 (Survey Certification)	[J1] Offline Model-Based Optimization: Comprehensive Review Minsu Kim, Jiayao Gu, Ye Yuan, <u>Taeyoung Yun</u> , Zixuan Liu, Yoshua Bengio, Can Chen Paper / Code
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TEACHING EXPERIENCES

2026	Teaching Assistant IE579: Game Theory and Multi-Agent Reinforcement Learning	KAIST
2025	Teaching Assistant IE343: Statistical Machine Learning IE437: Data-Driven Decision Making and Control	KAIST

2024	Teaching Assistant IE437: Data-Driven Decision Making and Control IE481: Manufacturing & Artificial Intelligence	KAIST
2022	Teaching Assistant MAS480: Introduction to Scientific Machine Learning	KAIST

ACADEMIC SERVICES

2025	Reviewer ICLR, AAMAS, AISTATS, ICML, KDD, TMLR, NeurIPS
2026	Reviewer AAAI, KDD, ICLR, AAMAS, AISTATS, TMLR, TPAMI, ICML, RLC

HONORS & AWARDS

2025	Qualcomm Innovation Fellowship Finalist Paper: Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation	Seoul, Korea
2025	Top Reviewer Honor for Top 8% Reviewers	NeurIPS
2021	Dean's List Honor for Top 2% Students	KAIST
2021	Excellence Award (2nd Place) Big Data Competition Hosted by NH	Seoul, Korea